

1. Set up the partial fraction decomposition of the following rational functions. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - 2x^2 + 3x - 4}{x(x-1)^2(x^2+9)^2} =$$

2. Evaluate the improper integral $\int_1^{\infty} \frac{\arctan x}{1+x^2} dx$ or show that it is divergent.

(Note: Answers without appropriate limit notations will result in loss of points.)